System Software - Expected Answer Sheet

February 2024 Mid Semester Paper

Course: CS306 - System Software
Exam: Mid Semester Examination - February 2024
Max Marks: 30

This write-up gives model student answers. For questions containing `OR`, both alternatives are included so the file is usable as a complete preparation sheet.

Q1

1. Explain the following with example (any two) [2 marks]

Below are all four, though any two are sufficient in the exam.

OPTAB

OPTAB is the operation table of the assembler. It stores mnemonic, machine opcode, instruction format and length.

Example:

```
MOVER -> opcode 04
ADD   -> opcode 01
STOP  -> opcode 00
```

Use: the assembler checks whether an opcode is valid and translates mnemonic to machine representation.

SYMTAB

SYMTAB is the symbol table. It stores user-defined symbols or labels along with their addresses.

Example:

```
LOOP -> 205
NEXT -> 210
```

Use: required for forward references and address substitution during Pass II.

LITAB

LITAB is the literal table. It stores all literals used in the program and the addresses assigned to them.

Example:

```
=5 -> 300
=1 -> 301
```

Use: literals are collected first and assigned storage during `LTORG` or `END`.

POOLTAB

POOLTAB stores the starting index of each literal pool in LITAB.

Example:

```
Pool 1 starts at literal 1
Pool 2 starts at literal 4
```

Use: helps assembler know which literals belong to the current pool when `LTORG` or `END` is encountered.

---

2. Language processing activities [2 marks]

Language processing activities convert a high-level source program into an executable program.

Main stages:

1. Preprocessor: expands macros, includes header files, removes comments.
2. Compiler: converts high-level language into assembly language or intermediate code.
3. Assembler: converts assembly language into machine code.
4. Linker: combines object modules and library routines.
5. Loader: places the executable in memory for execution.

Example flow:

```
C program -> Compiler -> Assembly -> Assembler -> Object code -> Linker -> Executable -> Loader -> Memory
```

OR: LEX program to remove comments from a C program [2 marks]

```
%{
#include <stdio.h>
%}

%%
"//".*                      ;
"/*"([^*]|\*+[^*/])*"*/"   ;
(.|\n)                       ECHO;
%%

int yywrap() {
    return 1;
}

int main() {
    yylex();
    return 0;
}
```

Explanation:

1. The first rule removes single-line comments.
2. The second rule removes block comments.
3. All other characters are echoed unchanged.

3. Macro for `A * B + C * D` with result in `AREG` [2 marks]

One suitable macro is:

```
MACRO
MULADD &A, &B, &C, &D, &R=AREG
MOVER &R, &A
MULT  &R, &B
MOVER BREG, &C
MULT  BREG, &D
ADD   &R, BREG
MEND
```

Example call:

```
MULADD X, Y, P, Q
```

Parameter types used:

1. `&A, &B, &C, &D` are positional parameters.
2. `&R=AREG` is a keyword parameter with default value `AREG`.

So the macro computes:

```
AREG <- A*B + C*D
```

---

4. Symbol Table at end of Pass I and Intermediate Code [4 marks]

Given source program:

```
START 200
MOVER AREG, =5
MOVEM AREG, M

L1   MOVER AREG, =2
     ORIGIN L1+3
     LTORG

NEXT ADD AREG, =1
     SUB BREG, =2
     BC LT, BACK
     LTORG

BACK EQU L1
     ORIGIN NEXT+5
     MULT CREG, =4
     STOP

X    DS 2
END
```

Important note

As printed in the paper, symbol `M` is referenced but never defined, while `X` is defined later and never used. Therefore, the exact Pass I answer for the printed program must show `M` as unresolved. If the intended last declaration was `M DS 2`, then `M` would receive address `214`.

Location counter trace

LC	Statement	Remark
-	`START 200`	initialize LC = 200
200	`MOVER AREG, =5`	literal `=5` seen
201	`MOVEM AREG, M`	symbol `M` referenced
202	`L1 MOVER AREG, =2`	`L1 = 202`, literal `=2` seen
-	`ORIGIN L1+3`	LC becomes 205
205	`LTORG`	assign `=5` at 205, `=2` at 206
207	`NEXT ADD AREG, =1`	`NEXT = 207`, literal `=1` seen
208	`SUB BREG, =2`	new literal occurrence `=2` seen
209	`BC LT, BACK`	`BACK` forward reference
210	`LTORG`	assign `=1` at 210, second `=2` at 211
-	`BACK EQU L1`	`BACK = 202`
-	`ORIGIN NEXT+5`	LC becomes 212
212	`MULT CREG, =4`	literal `=4` seen
213	`STOP`	
214	`X DS 2`	reserve 2 words, so next LC = 216
216	`END`	assign pending literal `=4` at 216

Symbol Table at end of Pass I

Index	Symbol	Address
1	`M`	undefined / unresolved
2	`L1`	202
3	`NEXT`	207
4	`BACK`	202
5	`X`	214

Intermediate Code

Using the usual notation:

```
AD = Assembler Directive
IS = Imperative Statement
DL = Declarative Statement
```

Registers:

```
AREG = 1, BREG = 2, CREG = 3
LT condition code = 1
```

One acceptable IC is:

LC	Statement	Intermediate Code
-	`START 200`	`(AD,01) (C,200)`
200	`MOVER AREG, =5`	`(IS,04) (1) (L,1)`
201	`MOVEM AREG, M`	`(IS,05) (1) (S,1)`
202	`L1 MOVER AREG, =2`	`(IS,04) (1) (L,2)`
-	`ORIGIN L1+3`	`(AD,03) (S,2) + (C,3)`
205	`LTORG`	`(AD,05)`
205	literal `=5`	`(DL,01) (C,5)`
206	literal `=2`	`(DL,01) (C,2)`
207	`NEXT ADD AREG, =1`	`(IS,01) (1) (L,3)`
208	`SUB BREG, =2`	`(IS,02) (2) (L,4)`
209	`BC LT, BACK`	`(IS,07) (1) (S,4)`
210	`LTORG`	`(AD,05)`
210	literal `=1`	`(DL,01) (C,1)`
211	literal `=2`	`(DL,01) (C,2)`
-	`BACK EQU L1`	`(AD,04) (S,4) (S,2)`
-	`ORIGIN NEXT+5`	`(AD,03) (S,3) + (C,5)`
212	`MULT CREG, =4`	`(IS,03) (3) (L,5)`
213	`STOP`	`(IS,00)`
214	`X DS 2`	`(DL,02) (C,2)`
216	`END`	`(AD,02)`
216	literal `=4`	`(DL,01) (C,4)`

Q2

1. FIRST, FOLLOW, predictive parsing table, and parse `(v+v)` [5 marks]

Grammar:

```
E -> T A
A -> + T A | - T A | e
T -> F B
B -> * F B | / F B | e
F -> - S | S
S -> v | ( E )
```

FIRST sets

```
FIRST(E) = { -, v, ( }
FIRST(A) = { +, -, e }
FIRST(T) = { -, v, ( }
FIRST(B) = { *, /, e }
FIRST(F) = { -, v, ( }
FIRST(S) = { v, ( }
```

FOLLOW sets

```
FOLLOW(E) = { ), $ }
FOLLOW(A) = { ), $ }
FOLLOW(T) = { +, -, ), $ }
FOLLOW(B) = { +, -, ), $ }
FOLLOW(F) = { *, /, +, -, ), $ }
FOLLOW(S) = { *, /, +, -, ), $ }
```

Predictive parsing table

Non-terminal	`v`	`(`	`)`	`+`	`-`	`*`	`/`	`$`
`E`	`E -> TA`	`E -> TA`	-	-	`E -> TA`	-	-	-
`A`	-	-	`A -> e`	`A -> +TA`	`A -> -TA`	-	-	`A -> e`
`T`	`T -> FB`	`T -> FB`	-	-	`T -> FB`	-	-	-
`B`	-	-	`B -> e`	`B -> e`	`B -> e`	`B -> *FB`	`B -> /FB`	`B -> e`
`F`	`F -> S`	`F -> S`	-	-	`F -> -S`	-	-	-
`S`	`S -> v`	`S -> (E)`	-	-	-	-	-	-

Since no cell contains more than one production, the grammar is LL(1).

Parsing action for `(v+v)`

Start stack: `$E`
Input: `(v+v)$`

Step	Stack	Input	Action
1	`$E`	`(v+v)$`	`E -> TA`
2	`$AT`	`(v+v)$`	`T -> FB`
3	`$ABF`	`(v+v)$`	`F -> S`
4	`$ABS`	`(v+v)$`	`S -> (E)`
5	`$AB)E(`	`(v+v)$`	match `(`

Step	Stack	Input	Action
6	`$AB)E`	`v+v)$`	`E -> TA`
7	`$AB)AT`	`v+v)$`	`T -> FB`
8	`$AB)ABF`	`v+v)$`	`F -> S`
9	`$AB)ABS`	`v+v)$`	`S -> v`
10	`$AB)ABv`	`v+v)$`	match `v`
11	`$AB)AB`	`+v)$`	`B -> e`
12	`$AB)A`	`+v)$`	`A -> +TA`
13	`$AB)AT+`	`+v)$`	match `+`
14	`$AB)AT`	`v)$`	`T -> FB`
15	`$AB)ABF`	`v)$`	`F -> S`
16	`$AB)ABS`	`v)$`	`S -> v`
17	`$AB)ABv`	`v)$`	match `v`
18	`$AB)AB`	`)$`	`B -> e`
19	`$AB)A`	`)$`	`A -> e`
20	`$AB)`	`)$`	match `)`
21	`$AB`	`$`	`B -> e`
22	`$A`	`$`	`A -> e`
23	`$`	`$`	accept

Hence the string `(v+v)` is accepted.

---

2. NFA and DFA for `a(ab)*a` [3 marks]

The regular expression means:

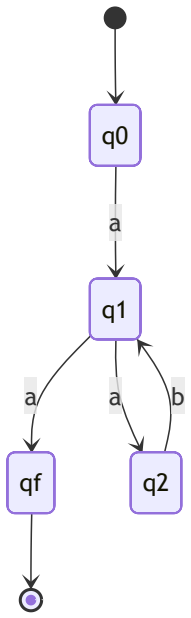
```
one initial a,
followed by zero or more repetitions of ab,
followed by one final a
```

So accepted strings are:

```
aa, aaba, aababa, aabababa, ...
```

NFA

One valid NFA is:



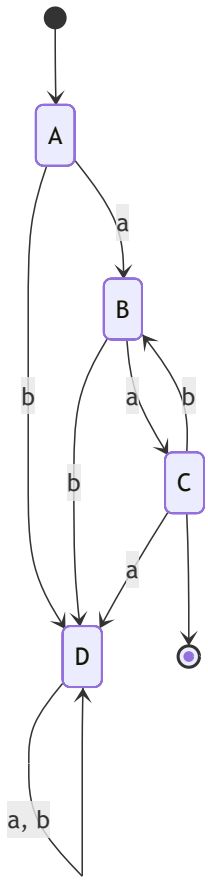
Here:

1. `q0` is start state.
2. `qf` is final state.
3. From `q1`, one `a` can directly finish the string.
4. Or it can move to `q2`, read `b`, come back to `q1`, and repeat `(ab)` any number of times.

DFA by subset construction

Let:

```
A = {q0}
B = {q1}
C = {q2, qf}
D = dead state
```



Transition table:

State	a	b	Final?
A	B	D	No
B	C	D	No
C	D	B	Yes
D	D	D	No

Thus `C` is the accepting DFA state because it contains `qf`.

OR: Transition diagrams for comments and identifiers in C [3 marks]

1. Comments in C

For single-line comment `//...`:

```
q0 --/--> q1 --/--> q2
q2 --any character except newline--> q2
q2 --newline--> accept
```

For block comment `/* ... */`:

```
q0 --/--> q1 --*--> q3
q3 --any char except *--> q3
q3 --*--> q4
q4 --/--> accept
q4 --*--> q4
q4 --other--> q3
```

2. Identifier in C

Rule: an identifier starts with letter or `_`, followed by letters, digits or `_`.

Transition diagram:

```
q0 --letter or _--> q1
q1 --letter or digit or _--> q1
```

`q1` is the accepting state.

---

3. Check whether the grammar is LL(1), construct predictive parse table, explain steps [4 marks]

Grammar:

```
S -> 0 | 1S2S3 | 1A3
A -> S | AS
```

FIRST sets

```
FIRST(S) = { 0, 1 }
FIRST(A) = { 0, 1 }
```

Observations

1. `S -> 1S2S3` and `S -> 1A3` have the same FIRST symbol `1`.
2. `A -> AS` is directly left recursive.

Therefore the grammar cannot be LL(1).

Predictive parse table (showing conflict)

Non-terminal	`0`	`1`	`2`	`3`	`$`
`S`	`S -> 0`	`S -> 1S2S3` and `S -> 1A3`	error	error	error
`A`	`A -> S` and `A -> AS`	`A -> S` and `A -> AS`	error	error	error

Since entries contain multiple productions, the parse table has conflicts.

Hence:

```
The grammar is NOT LL(1).
```

Steps used in constructing predictive parse table

1. Compute FIRST set of every non-terminal.
2. For each production `A -> alpha`, place it in `M[A, a]` for each `a` in `FIRST(alpha)`.
3. If `alpha` can derive epsilon, then place it in columns of `FOLLOW(A)` also.
4. If any table cell gets more than one production, the grammar is not LL(1).

This grammar fails because of common-prefix conflict and left recursion.

---

4. Eliminate left recursion and perform left factoring [2 marks]

Grammar:

```
S -> A
A -> Ad | Ae | aB | ac
B -> bBc | f
```

`A` has direct left recursion due to `Ad` and `Ae`.

Take:

```
alpha = d, e
beta  = aB, ac
```

After removing left recursion:

```
S  -> A
A  -> aB A' | ac A'
A' -> dA' | eA' | e
B  -> bBc | f
```

Now left factor `A`:

```
S  -> A
A  -> a X A'
X  -> B | c
A' -> dA' | eA' | e
B  -> bBc | f
```

This is the required grammar.

---

5. (a) Define tokens, lexemes, pattern [4 marks total with part b]

Token

A token is a category of lexical unit returned by the lexical analyser.

Examples: `id`, `num`, `if`, `+`, `;`

Lexeme

A lexeme is the actual sequence of characters in the source program that matches a token pattern.

Examples:

1. `count` is a lexeme of token `id`
2. `123` is a lexeme of token `num`

Pattern

A pattern is the rule that describes all lexemes belonging to a token.

Example:

```
identifier pattern = letter (letter | digit)*
```

---

5. (b) Total number of lexemes in the given code [continued]

Given code:

```c
#include
#include

int main()
{
    printf("%d + %d = %d",3,1,4); /*sum*/
    return 0;
}
```

Counting all program lexemes except whitespace/newlines, one acceptable count is:

```
1  #include
2  #include
3  int
4  main
5  (
6  )
7  {
8  printf
9  (
10 "%d + %d = %d"
11 ,
12 3
13 ,
14 1
15 ,
16 4
17 )
18 ;
19 /*sum*/
20 return
21 0
22 ;
23 }
```

Therefore, if the comment is counted as one lexeme, the total number of lexemes is:

```
23
```

If a teacher ignores comments during lexical counting, the answer becomes `22`. In most compiler-style counting, mentioning this assumption is the safest way to write the answer.

---

6. Check whether the grammar is ambiguous for string `aabbccdd` [2 marks]

Grammar:

```
S -> AB | C
A -> aAb | ab
B -> cBd | cd
C -> aCd | aDd
D -> bDc | bc
```

We show two different derivations for the same string `aabbccdd`.

Derivation 1: through `AB`

```
S => AB
  => aAb B
  => a(ab)b B
  => aabb B
  => aabb cBd
  => aabb c(cd)d
  => aabbccdd
```

Derivation 2: through `C`

```
S => C
  => aCd
  => aaDdd
  => aabDcdd
  => aabbccdd
```

Thus the same string has two distinct derivations.

Therefore, the grammar is:

```
ambiguous
```

Quick Revision Summary

Important answers from this paper:

1. Q2.1 grammar is LL(1), and `(v+v)` is accepted.
2. Q2.3 grammar is not LL(1).
3. Q2.6 grammar is ambiguous because `aabbccdd` has two derivations.
4. In Q1.4, the printed program leaves symbol `M` unresolved unless the paper intended `M DS 2` instead of `X DS 2`.